

Script-Controlled ACL – Restrict Record Access Based on Field Value

7. Final Project Report – Completed Answer

7.1 Introduction

Overview: A ServiceNow security lab demonstrating script-controlled ACLs for record-level access based on roles and field values.

Purpose: Demonstrate READ, CREATE, WRITE and DELETE record-level security.

7.2 Ideation Phase

The problem is unauthorized visibility or modification of institution records. The solution uses ACLs, role separation, a Branch condition and verification through impersonation.

7.3 Requirement Analysis

The solution requires the u_institution_details table, ECE/EEE/CSE Branch values, roles bb1–bb4 and four record ACLs. Fields are Student Roll Number, Student Name, Faculty Name, Branch, Email, Phone Number and Description.

7.4 Project Design

READ uses bb1, Branch = EEE and the supplied script. CREATE uses bb2, WRITE uses bb3, and DELETE uses bb4. No additional data condition is specified for the latter three.

7.5 Project Planning & Scheduling

The source lab specifies a 3-hour timer and milestones covering users/roles, table creation, ACL creation and verification. The separate planning answer supplies a proposed sprint mapping for the template.

7.6 Functional Testing

Verification	Expected outcome
bb1 user	Can view EEE branch records only.
User without required role	Cannot view records.
Admin	Can view all records regardless of branch.
bb1 + bb2	Can view EEE records and use New.
bb1 + bb2 + bb3	Can view and edit EEE records and use New.
bb1 + bb2 + bb3 + bb4	Can see, create, edit and delete EEE records.

7.7 Results

The target result is successful enforcement of record-level READ, CREATE, WRITE and DELETE ACLs. Actual screenshots are not present in the supplied source set and are therefore not fabricated.

7.8 Advantages

Advantage	Explanation
Record-level control	Access can depend on record data such as Branch.
Role separation	Different operations can require different roles.
Administrative access	The READ script explicitly allows admin users.

Advantage	Explanation
Verifiability	Impersonation provides a direct validation method.

7.9 Limitations

The source does not describe production-scale performance, automated regression testing, detailed audit requirements or deployment architecture. The Branch restriction is configured as a READ ACL data condition; the supplied script itself checks roles.

7.10 Conclusion

The project demonstrates how ServiceNow ACLs combine operation-level permissions, roles, record conditions and script logic to enforce access decisions.

7.11 Future Scope

Proposed extensions include broader branch policies, more granular roles, automated ACL regression tests, audit reporting and production deployment documentation.

7.12 Appendix

Reference source: Script-Controlled ACL – Restrict Record Access Based on Field Value. The source documents the ACL setup, verification and outcome.